

## Project Design Phase

### Problem – Solution Fit Template

|               |              |
|---------------|--------------|
| Date          | 28 June 2026 |
| Team ID       |              |
| Project Name  | House Rent   |
| Maximum Marks | 2 Marks      |

#### Problem – Solution Fit Template:

The Problem–Solution Fit for House Rent – House Rental Management System focuses on identifying the real challenges faced by renters and property owners and providing an effective web-based solution. The platform simplifies the complete house rental process by offering verified property listings, secure user authentication, online property management, booking requests, owner approval, and transparent communication. By addressing these challenges, House Rent ensures a reliable, secure, and user-friendly rental experience for all stakeholders.

#### Purpose:

- Simplify the house rental process through a centralized online platform.
- Enable renters to search, filter, and book verified rental properties easily.
- Help property owners upload, manage, and approve rental requests efficiently.
- Improve transparency with secure authentication and booking management.
- Reduce manual paperwork and communication delays.
- Provide a responsive and user-friendly interface for renters, owners, and administrators.

#### Template:

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Define CS, fit into CC                                                                                                                                                                                                                                                  |  <b>1. CUSTOMER SEGMENT(S)</b> <span style="float: right; background-color: #ffc107; padding: 2px 5px;">CS</span>                                                                                                                                                                                                                      |  <b>6. CUSTOMER CONSTRAINTS</b> <span style="float: right; background-color: #ffc107; padding: 2px 5px;">CC</span>                                                                                                                                                                                                                                                                                                          |  <b>5. AVAILABLE SOLUTIONS</b> <span style="float: right; background-color: #ffc107; padding: 2px 5px;">AS</span>                                                                                                                                                                                                                                                                                                          | Explore AS, differentiate                |
|                                                                                                                                                                                                                                                                         | <p>Who is your customer?</p> <p><b>In our project:</b></p> <ul style="list-style-type: none"> <li>• Renters / Tenants looking for houses</li> <li>• Property Owners who want to rent out their properties</li> <li>• Administrators who manage the platform</li> </ul>                                                                                                                                                    | <p>What constraints prevent your customers from taking action or limit their choices of solutions?</p> <p><b>In our project:</b></p> <ul style="list-style-type: none"> <li>• Difficulty in finding verified rental properties</li> <li>• Time-consuming manual search and paperwork</li> <li>• Lack of transparency in property details</li> <li>• Unreliable communication with owners</li> <li>• Secure payment and booking concerns</li> </ul>                                                             | <p>Which solutions are available to the customers when they face the problem or need to get the job done? What are the pros &amp; cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking</p> <p><b>In our project:</b></p> <ul style="list-style-type: none"> <li>• Broker / Agent Services (Expensive)</li> <li>• Classified Websites / Social Media (Unverified)</li> <li>• Offline Property Visits (Time-consuming)</li> <li>• Word of Mouth (Limited Options)</li> </ul> |                                          |
| Focus on J&P, tap into BE, understand RC                                                                                                                                                                                                                                |  <b>2. JOBS-TO-BE-DONE / PROBLEMS</b> <span style="float: right; background-color: #ffc107; padding: 2px 5px;">J&amp;P</span>                                                                                                                                                                                                          |  <b>9. PROBLEM ROOT CAUSE</b> <span style="float: right; background-color: #ffc107; padding: 2px 5px;">RC</span>                                                                                                                                                                                                                                                                                                            |  <b>7. BEHAVIOUR</b> <span style="float: right; background-color: #ffc107; padding: 2px 5px;">BE</span>                                                                                                                                                                                                                                                                                                                    | Focus on J&P, tap into BE, understand RC |
|                                                                                                                                                                                                                                                                         | <p>Which job-to-be-done (or problems) do your customers want to get done?</p> <p><b>In our project:</b></p> <ul style="list-style-type: none"> <li>• Find a suitable rental property</li> <li>• Get detailed and accurate property information</li> <li>• Book a property easily</li> <li>• Communicate with property owners</li> <li>• Make secure payments</li> <li>• Manage and track bookings / properties</li> </ul> | <p>What is the real reason that this problem exists? What is the back story behind the need to do this job?</p> <p><b>In our project:</b></p> <ul style="list-style-type: none"> <li>• Information is scattered across multiple platforms</li> <li>• No centralized system for verified listings and bookings</li> <li>• Manual communication leads to delays and confusion</li> <li>• Lack of secure and transparent booking process</li> <li>• No efficient way to manage properties and requests</li> </ul> | <p>What does your customer do to address the problem and get the job done? What do they think? What pros &amp; cons do they weigh? What motivates them?</p> <p><b>In our project:</b></p> <ul style="list-style-type: none"> <li>• Search properties on multiple platforms</li> <li>• Compare prices and contact owners/brokers</li> <li>• Visit properties physically</li> <li>• Negotiate and finalize terms</li> <li>• Manage bookings and payments manually</li> </ul>                                     |                                          |
| Identify strong TR & EM                                                                                                                                                                                                                                                 |  <b>3. TRIGGERS</b> <span style="float: right; background-color: #28a745; padding: 2px 5px;">TR</span>                                                                                                                                                                                                                                 |  <b>10. YOUR SOLUTION</b> <span style="float: right; background-color: #28a745; padding: 2px 5px;">SL</span>                                                                                                                                                                                                                                                                                                                |  <b>8. CHANNELS of BEHAVIOUR</b> <span style="float: right; background-color: #28a745; padding: 2px 5px;">CH</span>                                                                                                                                                                                                                                                                                                        |                                          |
|                                                                                                                                                                                                                                                                         | <p>What triggers customers to act?</p> <p><b>In our project:</b></p> <ul style="list-style-type: none"> <li>• Need to move to a new place</li> <li>• Change in job / location</li> <li>• Expiry of current lease</li> <li>• Better price or property availability</li> </ul>                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
|  <b>4. EMOTIONS: BEFORE / AFTER</b> <span style="float: right; background-color: #28a745; padding: 2px 5px;">EM</span>                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |
| <p>How do customers feel when they face a problem or a job and afterwards?</p> <p><b>In our project:</b></p> <ul style="list-style-type: none"> <li>• Before: Frustrated, confused, time-stressed, unsure</li> <li>• After: Relief, happy, secure, satisfied</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                          |

| Customer Problem                        | Current Situation                                              | House Rent Solution                                                             | Expected Outcome                                  |
|-----------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------|
| Finding genuine rental properties       | Renters rely on brokers or unverified online advertisements.   | Verified property listings with images, location, and complete details.         | Faster and more reliable property search.         |
| Manual booking process                  | Property bookings require phone calls and paperwork.           | Online booking system with owner approval workflow.                             | Convenient and efficient booking process.         |
| Difficult property management           | Owners manually manage property listings and tenant requests.  | Owner dashboard for uploading, updating, and managing properties.               | Simplified property management.                   |
| Lack of secure user access              | Unauthorized users may access sensitive information.           | Secure role-based authentication for Admin, Owner, and Renter.                  | Improved data security and controlled access.     |
| Poor communication and booking tracking | Users cannot easily monitor booking requests or approvals.     | Real-time booking status (Pending, Approved, Rejected) and transaction history. | Better transparency and user satisfaction.        |
| Time-consuming rental process           | Searching and finalizing a rental property takes several days. | Advanced search filters, responsive interface, and centralized platform.        | Reduced rental time and improved user experience. |